



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Third Semester : Fall 2023

Subject: Calculus and Analytical Geometry

Paper: MS-251 N

Roll No.
Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Define the following:

(6x5=30)

- I. Find domain and range of the function $y = \frac{x^2+x-2}{x^2-x-2}$?
- II. Evaluate the integral $\int \cos 2t \sin^5 2t \, dt$?
- III. Find parametric equations of the line through $(2, -1, 5)$ that is parallel to $\langle -1, 2, 7 \rangle$.
- IV. Find second order derivative of $f(x) = (1 + 3x)^8$?
- V. Show that $y = \cos x$ and $y = \sin x$ are the solutions to $y'' + y = 0$.
- VI. Find an equation of the plane through the points $P_1(1, 2, -1)$, $P_2(2, 3, 1)$, and $P_3(3, -1, 2)$.

Q.2. Answer the following questions.

(3x10=30)

- I. Find the intervals on which $g(x) = -x^3 + 12x + 5$, $-3 \leq x \leq 3$ is increasing and decreasing. Where does the function assume the extreme values and what are these values?
- II. a) Evaluate the integral $\int r^4 \left(7 - \frac{r^5}{10}\right)^3 dr$?
b) Evaluate the integral $\int_0^{\pi} \frac{1}{2} (\cos x + |\cos x|) dx$?
- III. Find parametric equations for the line in which the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ intersect?